

Stellar Subdivision Schemes and Pointerless Maubach Bisection on Balanced Seeds

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Abstract

We give a self-contained combinatorial account of Maubach’s newest-vertex bisection as a *stellar subdivision scheme*: an invariant on ordered simplicial meshes that holds at every level of the hierarchy, together with a subdivision algorithm that provably preserves it. We then turn to the initial mesh. Strong compatibility of a seed is equivalent to the seed being balanced, i.e. properly $(d+1)$ -colorable, a fact implicit in Traxler’s work and in recent coloring-based initializations. We use it to connect bisection with the classes of triangulations of closed manifolds studied by Dynnikov, Novikov and Gaifullin. In particular, Gaifullin’s 15-vertex triangulation of $\mathbb{CP}^2$ is a strongly compatible seed with the fewest possible vertices. Balance is also what makes Atalay and Mount’s pointerless LPT codes portable beyond the cube. On a balanced seed, a precomputed table plus one integer comparison decides whether a facet lies on the seed boundary, and the neighbor across it carries the same code in the adjacent seed cell. This yields a conforming, graded, pointerless Maubach bisection over any balanced seed, with no floating point. As an example, we triangulate smooth plane curves intrinsically inside $\mathbb{CP}^2$.

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1 Introduction

Newest-vertex bisection in the form given by Maubach [32] and Traxler [43] is the standard way to refine simplicial meshes locally in arbitrary dimension d . Its refinement decisions depend only on combinatorial data: a vertex order in each cell and the cell’s level in the hierarchy. It keeps meshes conforming, produces finitely many similarity classes when new vertices are placed at midpoints [32, 4], and satisfies the closure estimates behind optimal adaptive finite elements [10, 42]. Its well-known weakness is the initial mesh. The vertex orders of the initial cells must be *compatible*, and for $d \geq 3$ not every mesh admits compatible orders. A substantial literature addresses how to repair this [27, 42, 1, 8, 17].

This paper studies Maubach bisection from a purely combinatorial standpoint, with two goals in mind: to state and prove its correctness through local stellar operations alone, and to run it pointerlessly on seeds more general than the Kuhn cube. The original motivation was geometric: to triangulate curves inside $\mathbb{CP}^2$, whose natural triangulations are neither cubes nor domains in $\mathbb{R}^d$.

Stellar subdivision schemes (Sections 3 and 4). Following the author’s thesis [36], a *subdivision scheme* is an invariant on ordered simplicial meshes together with an algorithm that propagates ordered stellar subdivisions until a requested cell can be split without violating the invariant. We give complete proofs for two schemes, Mitchell’s (split a facet) and Maubach’s (split an edge), which share the same three-lemma structure. For Maubach’s scheme, the invariant is *hereditary*: it holds at every level, not only on the initial mesh as in the conditions of Traxler and Stevenson. We prove one direction of its relation to Stevenson’s matching condition.

Balanced seeds (Section 5). On a conforming mesh, strong compatibility (every shared facet sits at the same local index on both sides) is equivalent to the mesh being *balanced*. This equivalence is implicit in Traxler’s work [43] and explicit, in the language of vertex colorings, in the recent initialization of Diening, Gehring and Storn [17]; we include a short proof. Our contribution is to connect it with the classes of triangulations of closed manifolds studied by Dynnikov and Novikov and by Gaifullin [23], which the bisection literature does not consider. For simply connected manifolds these classes coincide, so a seed is strongly compatible if and only if every codimension-2 face has even degree. Gaifullin proved that his 15-vertex triangulation of $\mathbb{CP}^2$ is vertex-minimal among balanced triangulations of $\mathbb{CP}^2$. It follows that it is a strongly compatible seed of $\mathbb{CP}^2$ with the fewest possible vertices. Kühnel’s 9-vertex $\mathbb{CP}_9^2$ is not balanced.

Pointerless codes on balanced seeds (Section 6). Atalay and Mount’s LPT codes [5] represent a Maubach hierarchy over a Kuhn-triangulated cube with no pointers. The geometry

and the neighbors of a cell are computed from a small integer code. We show that balance is all that is needed to use these codes over an arbitrary seed:

- whether a facet lies on the seed boundary is decided by a table indexed by the code’s level, signed permutation and facet, followed by one comparison of orthant coordinates (Proposition 6.1);
- the neighbor across a seed-boundary facet is the same code in the adjacent seed cell (Proposition 6.2).

Every neighbor query is therefore resolved combinatorially. We verify the resulting procedure exactly on five million queries, and compare its memory use with that of explicit incidence structures.

Example (Section 7). We triangulate smooth plane curves directly inside $\mathbb{CP}^2$. The seed is Gaifullin’s triangulation, refinement is pointerless Maubach bisection along Fubini–Study geodesics, and the curve is followed by a continuation driven only by combinatorial neighbor queries.

Contributions.

1. Complete, self-contained proofs of correctness for the Mitchell and Maubach stellar subdivision schemes, with a hereditary invariant, and one direction of its relation to Stevenson’s matching condition.
2. The connection between strongly compatible seeds and the classes of balanced, chess-colorable and even triangulations of closed manifolds. As a consequence, Gaifullin’s triangulation is a vertex-minimal strongly compatible seed of $\mathbb{CP}^2$.
3. An extension of LPT codes to arbitrary balanced seeds with a fully combinatorial neighbor procedure, an exact verification, and a memory comparison with explicit incidence structures.

2 Related work

Bisection and initial meshes. Longest-edge bisection goes back to Rivara [40, 41]. Newest-vertex bisection was introduced in 2D by Mitchell [37, 38], extended to 3D by Bänsch [6] and Kossaczky [27], and to arbitrary dimension by Maubach [32] and Traxler [43]. Arnold, Mukherjee and Pouly [4] gave a marked variant for tetrahedra that accepts any initial mesh. Binev, Dahmen and DeVore [10] and Stevenson [42] proved the closure estimates behind the optimality of adaptive finite elements. All of these methods constrain the initial mesh:

- Traxler’s *reflected* condition requires the ordered vertex lists of neighbors to differ in exactly one position; on a simply connected domain it holds iff every interior $(d-2)$ -face has even degree [43].
- Stevenson’s weaker *matching-neighbour* condition is necessary and sufficient for all uniform refinements to be conforming. Whether every 3D mesh admits it is open; following Kossaczky, Stevenson pre-refines every simplex into $\frac{1}{2}(d+1)!$ pieces [42].
- Alkämper, Gaspoz and Klöfkorn [1] give an $O(n)$ renumbering that guarantees termination.
- Belda-Ferrín et al. [7, 9, 8] perform d marked-bisection stages before switching to Maubach.
- Diening, Gehring and Storn [17] initialize Maubach’s routine from a proper $(N+1)$ -coloring of the vertices, $N \geq d$, computed greedily in linear time, and prove shape regularity and the closure estimate. For $N = d$ their condition is Traxler’s.

All of the above concern triangulations of domains in $\mathbb{R}^d$. We add the hereditary formulation of the invariant, and seeds that triangulate closed manifolds.

Balanced and chess-colorable triangulations. A d -complex is *balanced* if its graph is $(d+1)$ -colorable. Joswig [26] characterized balanced complexes through their group of projectivities. Dynnikov and Novikov studied triangulations with regular vertex colorings, chess colorings of top simplices, and even codimension-2 degrees in connection with discrete connections and discrete complex analysis; Gaifullin [23, §2] summarizes the relations between these classes and constructs a vertex-minimal balanced triangulation of $\mathbb{CP}^2$. Izmestiev, Klee and Novik [25] introduced cross-flips, the balanced analogue of bistellar moves, and Venturello [46] found small balanced triangulations of several manifolds. Stellar operations go back to Alexander and Newman [31]; in geometry processing, Velho’s stellar subdivision grammars [44] and 4–8 meshes [45] describe adaptive surface refinement by stellar operations, and 4–8 refinement is newest-vertex bisection in two dimensions.

Pointerless hierarchical meshes. Several representations encode a simplex of a regular bisection hierarchy by a label instead of pointers: Hebert [24]; Evans, Kirkpatrick and Townsend [22]; Lee, De Floriani and Samet [30]; Maubach’s own neighbor location [33]; and Atalay and Mount’s LPT codes [5] in arbitrary dimension, which we previously used for adaptive cellular structures in 3D printing [20]. Weiss and De Floriani use the star of the bisection edge (a *diamond*) as primitive [48]; Dupuy’s concurrent binary trees implement longest-edge bisection on the GPU [18]; Boissonnat, Kachanovich and Wintraecken encode Freudenthal–Kuhn simplices compactly, without adaptivity [12]. Architecturally closest to our construction are the forests of p4est [15] and t8code [14]: an arbitrary coarse mesh refined tree by tree, with a per-tree linear code and an inter-tree connectivity table. Those forests are non-conforming (hanging nodes, red refinement, $d \leq 3$); ours is a conforming, graded bisection forest in arbitrary dimension.

Implicit curves and surfaces. The example of Section 7 relates to piecewise-linear approximation of implicit manifolds over Freudenthal–Kuhn triangulations [3, 2], to its certified, uniform-grid variant by cell-to-cell tracing [12, 13], to the extraction of zero sets of functions $\mathbb{C}^2 \rightarrow \mathbb{C}$ on uniform grids of $\mathbb{R}^4$ [47], and to numerical and visual approaches to compact Riemann surfaces through branched coverings and charts [16, 11, 39, 28].

3 Ordered meshes and stellar subdivision schemes

Adaptive refinement has a geometric side (where new vertices go, how cell shapes evolve) and a combinatorial side (which simplex to split next, and which additional splits keep the mesh conforming). Newest-vertex bisection is attractive precisely because its decisions can be made from combinatorial data alone, which makes it usable on meshes whose geometry is unusual, such as a triangulation of $\mathbb{CP}^2$ realized with Fubini–Study geodesics. This section isolates that combinatorial layer. A refinement *scheme* is an invariant on ordered meshes together with an algorithm that preserves it; correctness then reduces to checking how one stellar operation affects pairs of adjacent cells. We illustrate the framework on Mitchell’s scheme, whose proof has the same structure as Maubach’s but fewer cases.

Throughout, d denotes the dimension of the mesh.

3.1 Ordered simplicial complexes and conforming meshes

Definition 3.1. An *ordered simplicial complex* K on a finite vertex set V is an abstract simplicial complex with a partial order on V that is total on the vertices of every simplex. We write

$\sigma = \langle v_0, \dots, v_k \rangle$ with $v_0 < \dots < v_k$ and $\sigma[p] = v_p$. The *face operator* ∂_i deletes the i -th vertex, and $\sigma[p_0, \dots, p_q] := \langle v_{p_0}, \dots, v_{p_q} \rangle$ for $p_0 < \dots < p_q$.

Equalities such as $\partial_i \sigma = \partial_j \omega$ are equalities of *ordered* simplices; this will matter in Remark 5.3. The d -simplices of a d -complex are its *cells* and its $(d-1)$ -simplices its *facets*. Two cells are *adjacent* if they share a facet. K is a *pseudomanifold* if it is pure and every facet lies in at most two cells. It is *strongly connected* if any two cells are joined by a chain of adjacent cells.

Definition 3.2 ([36]). A strongly connected pseudomanifold is a *conforming mesh* if the star $\text{st}(v, K)$ of every vertex is strongly connected.

Every combinatorial manifold of dimension $d \geq 2$ is a conforming mesh. In a conforming mesh the star of every simplex ϵ is strongly connected. Moreover, two cells of $\text{st}(\epsilon, K)$ that share a facet share a facet containing ϵ , since both contain ϵ . Hence $\text{st}(\epsilon, K)$ can be traversed by crossing only facets that contain ϵ .

3.2 Ordered stellar subdivisions

Let $\delta \in K$ and let v be a new vertex. The *stellar subdivision* $K(\delta, v)$ replaces $\text{st}(\delta, K)$ by the cone from v over $\partial\delta \star \text{link}(\delta, K)$. It is an *ordered stellar subdivision* $K(\delta, v)_l$ if the new vertex is placed at position l in every new cell. A cell $\sigma \supseteq \delta$ with $\delta = \sigma[p_0, \dots, p_k]$ is replaced by $k+1$ children $c_0\sigma, \dots, c_k\sigma$, one for each facet of δ ; the child $c_q\sigma$ is obtained from σ by deleting $\sigma[p_{k-q}]$ and inserting v at position l . Hence

$$\partial_l c_q \sigma = \partial_{p_{k-q}} \sigma. \quad (1)$$

A sequence of ordered stellar subdivisions of an initial mesh K_0 induces a hierarchy: $\text{level}(\sigma) = 0$ for $\sigma \in K_0$ and $\text{level}(c_q \sigma) = \text{level}(\sigma) + 1$. A cell of positive level has a unique parent, and a unique *weld vertex* $\text{weld}(\sigma)$: the vertex inserted by the subdivision that created it.

3.3 Subdivision schemes

Definition 3.3 ([36]). A *stellar subdivision scheme* is a pair $(\mathcal{I}, \text{Subdivide})$ where $\mathcal{I}$ is a set of invariants on conforming meshes, stated in terms of vertex order, adjacency and level only, and $\text{Subdivide}(K, \sigma)$ applies a finite sequence of ordered stellar subdivisions to K , the last of which subdivides a face of σ (the *split face* of σ), such that every intermediate mesh satisfies $\mathcal{I}$.

A basic invariant bounds the level jump across facets:

$$\partial_i \sigma = \partial_j \omega \implies |\text{level}(\sigma) - \text{level}(\omega)| \leq 1. \quad (\text{OneLevel})$$

All invariants below are conditions on *pairs of adjacent cells*. To show that a stellar subdivision preserves them, it therefore suffices to check the adjacent pairs involving at least one new cell.

We say that Subdivide is *consistent* if every stellar subdivision (δ, v) it performs splits cells that all have split face δ ; *alternating* if $\text{weld}(\sigma)$ never belongs to the split face of σ ; and *level-homogeneous* if every stellar subdivision it performs splits cells of a single level. (The thesis calls the last property “balanced”; we rename it to avoid a clash with balanced complexes.)

3.4 Mitchell’s scheme

Mitchell’s scheme splits the facet opposite the first vertex: $\text{split}(\sigma) = \partial_0 \sigma$, with the new vertex inserted at position 0. For $\omega = \langle x, v_1, \dots, v_d \rangle$ the children of $(\partial_0 \omega, v)_0$ are, by (1),

$$c_q \omega = \langle v, x, v_1, \dots, \widehat{v_{d-q}}, \dots, v_d \rangle, \quad q = 0, \dots, d-1.$$

Algorithm 1 MITCHELLSUBDIVIDE(K, σ)

1: $\omega \leftarrow$ the cell adjacent to σ across $\partial_0\sigma$ (if any)
2: **if** ω exists and $\text{level}(\omega) < \text{level}(\sigma)$ **then**
3: MITCHELLSUBDIVIDE(K, ω); $\omega \leftarrow$ the cell adjacent to σ across $\partial_0\sigma$
4: **end if**
5: apply $(\partial_0\sigma, v)_0$ to σ and ω

Definition 3.4 (Mitchell invariants). K satisfies the *Mitchell invariants* if it satisfies (OneLevel) and, for adjacent cells with $\partial_i\sigma = \partial_j\omega$,

$$\text{level}(\sigma) = \text{level}(\omega) \implies i = 0 = j \text{ or } i \neq 0 \neq j, \quad (\text{MitchellA})$$

$$\text{level}(\sigma) = \text{level}(\omega) + 1 \implies i = 0 \neq j. \quad (\text{MitchellB})$$

Every mesh satisfies these invariants after the cone $(\sigma, v_\sigma)_0$ is applied to each of its cells: all new cells have level 0, and two adjacent new cells share a facet that avoids both cone vertices, so $i \neq 0 \neq j$.

Lemma 3.5 (direction). *Under the Mitchell invariants, if $\partial_i\sigma = \partial_j\omega$ then $\text{level}(\omega) \leq \text{level}(\sigma)$ when $i = 0$, and $\text{level}(\omega) \geq \text{level}(\sigma)$ when $i \neq 0$.*

Proof. If $\text{level}(\omega) > \text{level}(\sigma)$, then $\text{level}(\omega) = \text{level}(\sigma) + 1$ by (OneLevel), and (MitchellB) applied to the pair (ω, σ) gives $j = 0 \neq i$. If $\text{level}(\omega) < \text{level}(\sigma)$, (MitchellB) gives $i = 0$. $\square$

Lemma 3.6 (children). *Any two children of $(\partial_0\omega, v)_0$ are adjacent, and their common facet is obtained from each of them by deleting a vertex at a position ≥ 2 . In particular every such pair satisfies (MitchellA).*

Proof. The children form the cone $v \star \partial\delta \star x$ over the boundary of the facet $\delta = \partial_0\omega$. The children $c_q\omega$ and $c_r\omega$ omit two different vertices v_a, v_b of δ and share the remaining d vertices, which form a facet of both. In $c_q\omega$ this facet is obtained by deleting v_b , which sits after v and x , hence at a position ≥ 2 ; symmetrically in $c_r\omega$. (The thesis lists only the pairs omitting consecutive vertices, which are all the pairs when $d \leq 3$.) $\square$

Lemma 3.7 (two cells). *Let σ, ω be adjacent, $\partial_i\sigma = \partial_j\omega$, with $0 \leq \text{level}(\sigma) - \text{level}(\omega) \leq 1$, satisfying the Mitchell invariants. Subdivide ω by $(\partial_0\omega, v)_0$, and also σ if $\partial_0\sigma = \partial_0\omega$. Then all adjacent pairs of new cells, and of new cells with σ , satisfy the invariants.*

Proof. By (MitchellA) and (MitchellB), exactly three cases occur.

- (1) $\text{level}(\sigma) = \text{level}(\omega)$, $i = j = 0$. Both cells are split across their common facet. Pairs of children of the same cell satisfy (MitchellA) by Lemma 3.6. Since $\partial_0\sigma = \partial_0\omega$ as ordered simplices, $c_q\sigma$ and $c_q\omega$ differ only in position 1, so $\partial_1c_q\sigma = \partial_1c_q\omega$: (MitchellA). A child of σ and a child of ω with different indices share only $d - 1$ vertices.
- (2) $\text{level}(\sigma) = \text{level}(\omega)$, $i \neq 0 \neq j$. Only ω is split. The facet $\partial_j\omega$ contains $x = \omega[0]$ and omits $v_j = \omega[j]$, so it is $\partial_0c_{d-j}\omega$. The pair $(c_{d-j}\omega, \sigma)$ has levels $\text{level}(\sigma) + 1, \text{level}(\sigma)$ and indices $0 \neq i$: (MitchellB).
- (3) $\text{level}(\sigma) = \text{level}(\omega) + 1$, $i = 0 \neq j$. As in (2), $\partial_0\sigma = \partial_j\omega = \partial_0c_{d-j}\omega$, now at equal levels with indices $0, 0$: (MitchellA).

In all cases the level difference is at most 1, so (OneLevel) holds. $\square$

Theorem 3.8. *Let K satisfy the Mitchell invariants. For every cell σ , Algorithm 1 terminates, only splits cells of level at most $\text{level}(\sigma)$, and produces a mesh satisfying the Mitchell invariants. It is consistent, alternating and level-homogeneous.*

Proof. By induction on $\text{level}(\sigma)$. By Lemma 3.5 with $i = 0$, the neighbor ω across $\partial_0\sigma$ has $\text{level}(\omega) \leq \text{level}(\sigma)$. If $\text{level}(\omega) < \text{level}(\sigma)$, the recursive call has a smaller argument level; by the induction hypothesis it terminates, splits only cells of level at most $\text{level}(\omega)$ (so not σ), and preserves the invariants. By (MitchellB), $\partial_0\sigma = \partial_j\omega$ with $j \neq 0$, so after ω is split this facet belongs to the child $c_{d-j}\omega$, of level $\text{level}(\sigma)$, as its facet ∂_0 (case (3) of Lemma 3.7). In either case, when line 5 is reached, σ and the cell ω across $\partial_0\sigma$ have the same level and $\partial_0\sigma = \partial_0\omega$. These two cells are the whole star of the facet $\partial_0\sigma$, both have it as split face (consistency), and both have the same level (level homogeneity). Lemma 3.7, applied to every adjacent pair involving a new cell, shows that the invariants are preserved. Finally, the weld vertex v of a child sits at position 0 and $\text{split} = \partial_0$ omits position 0, so the scheme is alternating. $\square$

4 Maubach's scheme

In dimension 2 a facet is an edge, and Mitchell's scheme is exactly newest-vertex bisection. In higher dimensions a facet split is no longer a bisection. Maubach's scheme splits an edge; on Kuhn's triangulation of the cube it reproduces longest-edge bisection with finitely many similarity classes [32]. This section describes it as a stellar subdivision scheme: which invariant on ordered meshes guarantees that the edge chosen by one cell is also the edge chosen by every neighbor that contains it? The resulting proof of conformity refers only to vertex orders and levels, and applies to any mesh that satisfies the invariant, whatever its geometry.

For a cell σ of a d -mesh set

$$\text{type}(\sigma) := d - (\text{level}(\sigma) \bmod d) \in \{1, \dots, d\}, \quad \text{split}(\sigma) := \sigma[0, \text{type}(\sigma)],$$

and split $\text{split}(\sigma)$ with the new vertex at position $l = \text{type}(\sigma)$. By (1), the children of $\sigma = \langle v_0, \dots, v_d \rangle$ are

$$c_0\sigma = \langle v_0, \dots, v_{l-1}, v, v_{l+1}, \dots, v_d \rangle, \quad c_1\sigma = \langle v_1, \dots, v_l, v, v_{l+1}, \dots, v_d \rangle,$$

which is exactly the rule of [32] and of [17, Alg. 1]. Since $\text{type} \neq 0$, the facet $\partial_p\sigma$ contains $\text{split}(\sigma)$ iff $p \notin \{0, \text{type}(\sigma)\}$.

Definition 4.1 (Maubach invariants). K satisfies the *Maubach invariants* if it satisfies (OneLevel) and, for adjacent cells with $\partial_i\sigma = \partial_j\omega$,

$$\text{level}(\sigma) = \text{level}(\omega) \implies i = j \text{ or } \{i, j\} = \{0, \text{type}(\omega)\}, \quad (\text{MaubachA})$$

$$\text{level}(\sigma) = \text{level}(\omega) + 1 \implies i = \text{type}(\omega) \text{ and } j \in \{0, \text{type}(\omega)\}. \quad (\text{MaubachB})$$

K is *strongly compatible* if all its cells have level 0 and $\partial_i\sigma = \partial_j\omega$ implies $i = j$; such a mesh satisfies the Maubach invariants.

Lemma 4.2 (direction). *Under the Maubach invariants, if $\partial_i\sigma = \partial_j\omega$ then $\text{level}(\omega) \leq \text{level}(\sigma)$ when $i \notin \{0, \text{type}(\sigma)\}$, and $\text{level}(\omega) \geq \text{level}(\sigma)$ when $i \in \{0, \text{type}(\sigma)\}$. Moreover, if $\text{level}(\omega) = \text{level}(\sigma)$ and $i \notin \{0, \text{type}(\sigma)\}$, then $i = j$ and $\text{split}(\omega) = \text{split}(\sigma)$.*

Proof. If $\text{level}(\omega) > \text{level}(\sigma)$, then (OneLevel) and (MaubachB) for the pair (ω, σ) give $i \in \{0, \text{type}(\sigma)\}$. If $\text{level}(\omega) < \text{level}(\sigma)$, (MaubachB) gives $i = \text{type}(\omega)$, and $\text{type}(\omega) \notin \{0, \text{type}(\sigma)\}$ because consecutive levels have different types. If the levels are equal and $i \notin \{0, \text{type}(\sigma)\}$, then $\{i, j\} \neq \{0, \text{type}\}$, so (MaubachA) gives $i = j$; the two cells then differ only at position i , which is outside $\{0, \text{type}\}$, so their split edges coincide. $\square$

Lemma 4.3 (children). *For a cell ω and $l = \text{type}(\omega)$, the children satisfy $\partial_0 c_0 \omega = \partial_{l-1} c_1 \omega$, and this pair satisfies (MaubachA).*

Proof. Both sides equal $\langle v_1, \dots, v_{l-1}, v, v_{l+1}, \dots, v_d \rangle$. The children have equal level and type $l' = \text{type}(c_q \omega)$, which is $l - 1$ if $l > 1$ and d if $l = 1$. If $l > 1$ the indices are $\{0, l - 1\} = \{0, l'\}$; if $l = 1$ both indices are 0. $\square$

Lemma 4.4 (two cells). *Let σ, ω be adjacent, $\partial_i \sigma = \partial_j \omega$, with $0 \leq \text{level}(\sigma) - \text{level}(\omega) \leq 1$, satisfying the Maubach invariants, and let $l = \text{type}(\omega)$. Split ω at $\text{split}(\omega)$, and also σ if it contains $\text{split}(\omega)$. Then every adjacent pair formed by a child of ω and σ or a child of σ satisfies the Maubach invariants.*

Proof. We list all cases allowed by (MaubachA) and (MaubachB), writing $\omega = \langle v_0, \dots, v_d \rangle$. In each case we exhibit the facet identity. When the two cells have different levels, the coarser one is σ , of type l , and we check (MaubachB): the finer index must be l and the coarser index must lie in $\{0, l\}$.

case	levels, indices	new identity	invariant
1	equal, $i = j = 0$	$\partial_0 \sigma = \partial_l c_1 \omega$	(B): finer l , coarser 0
2	equal, $i = j = l$	$\partial_l \sigma = \partial_l c_0 \omega$	(B): finer l , coarser l
3	equal, $i = j = k \notin \{0, l\}$	$\partial_k c_0 \sigma = \partial_k c_0 \omega, \partial_{k'} c_1 \sigma = \partial_{k'} c_1 \omega$	(A): $i = j$
4	equal, $i = 0, j = l$	$\partial_0 \sigma = \partial_l c_0 \omega$	(B): finer l , coarser 0
5	equal, $i = l, j = 0$	$\partial_l \sigma = \partial_l c_1 \omega$	(B): finer l , coarser l
6	+1, $i = l, j = 0$	$\partial_l \sigma = \partial_l c_1 \omega$	(A): $i = j$
7	+1, $i = l, j = l$	$\partial_l \sigma = \partial_l c_0 \omega$	(A): $i = j$

In case 3, $k' = k - 1$ if $k < l$ and $k' = k$ if $k > l$.

Completeness. At equal levels, if $j \notin \{0, l\}$ then the common facet contains $\text{split}(\omega)$, so σ is split as well and $i = j$ by Lemma 4.2 (case 3). Otherwise $j \in \{0, l\}$, and (MaubachA) leaves cases 1, 2, 4 and 5. At levels differing by one, (MaubachB) for (σ, ω) gives $i = l, j \in \{0, l\}$ (cases 6 and 7).

Identities. $\partial_0 \omega = \langle v_1, \dots, v_d \rangle$ is $c_1 \omega$ with v (at position l) deleted, and $\partial_l \omega = \langle v_0, \dots, \widehat{v}_l, \dots, v_d \rangle$ is $c_0 \omega$ with v deleted. This gives cases 1, 2 and 4–7.

In case 3, σ and ω differ only at position k and have the same split edge. Their children $c_0 \sigma, c_0 \omega$ still differ only at position k . Their children $c_1 \sigma, c_1 \omega$ differ only at position $k - 1$ if $k < l$ (entries $1, \dots, l$ shift down by one) or at position k if $k > l$. A child of σ and a child of ω with different indices share at most $d - 1$ vertices, so these are the only new adjacencies.

Levels. In cases 1, 2, 4 and 5 the child has level $\text{level}(\sigma) + 1$; in cases 6 and 7 the child's level equals $\text{level}(\sigma)$. (OneLevel) holds in all cases. $\square$

Theorem 4.5. *Let K be a conforming mesh satisfying the Maubach invariants, and σ a cell. Then Algorithm 2 terminates, only splits cells of level at most $\text{level}(\sigma)$, and produces a conforming mesh satisfying the Maubach invariants. It is consistent, alternating and level-homogeneous.*

Proof. By induction on $\ell = \text{level}(\sigma)$. Let $\epsilon = \text{split}(\sigma)$ and $k = \text{type}(\sigma)$. We claim that every cell pushed onto Q has level ℓ , contains ϵ and has split edge ϵ , and that no cell of level $\geq \ell$ is split before the final line.

Let c be such a cell, and $i \notin \{0, \text{type}(c)\}$; the facet $\partial_i c$ contains ϵ . By Lemma 4.2, the cell c_a across it has level $\leq \ell$.

- If $\text{level}(c_a) = \ell$, then Lemma 4.2 gives $\text{split}(c_a) = \epsilon$.
- If $\text{level}(c_a) = \ell - 1$, the recursive call has argument level $\ell - 1$. By the induction hypothesis it terminates, preserves the invariants, and splits only cells of level $\leq \ell - 1$. In particular it

Algorithm 2 MAUBACHSUBDIVIDE(K, σ) [36]

```
1:  $k \leftarrow \text{type}(\sigma)$ ;  $\epsilon \leftarrow \sigma[0, k]$ ;  $st \leftarrow []$ ;  $Q \leftarrow (\sigma)$ 
2: while  $Q \neq \emptyset$  do
3:    $c \leftarrow \text{pop}(Q)$ 
4:   if  $c$  not visited then
5:     mark  $c$  visited; append  $c$  to  $st$ 
6:     for  $i \in \{1, \dots, d\} \setminus \{\text{type}(c)\}$  do  $\triangleright$  facets of  $c$  containing  $\epsilon$ 
7:        $c_a \leftarrow$  the cell adjacent to  $c$  across  $\partial_i c$  (skip if none)
8:       if  $\text{level}(c_a) < \text{level}(c)$  then
9:         MAUBACHSUBDIVIDE( $K, c_a$ );  $c_a \leftarrow$  the cell adjacent to  $c$  across  $\partial_i c$ 
10:      end if
11:      if  $c_a$  not visited then push  $c_a$  onto  $Q$ 
12:    end for
13:  end if
14: end while
15: split  $\epsilon$ , with the new vertex at position  $k$ , in every cell of  $st$ 
```

splits no visited or queued cell, so c is still a cell of the mesh. By (MaubachB), $\partial_i c = \partial_j c_a$ with $j \in \{0, \text{type}(c_a)\}$; this facet does not contain $\text{split}(c_a)$. When c_a is split, the facet passes whole to a child of level ℓ , by case 6 or 7 of Lemma 4.4, and cells of level ℓ are never split during the recursion. The cell across $\partial_i c$ is now of level ℓ , and the first case applies.

This proves the claim. Each recursive call strictly lowers the level of its argument, and each call visits each cell at most once, so the algorithm terminates.

When the loop ends, st consists of cells of level ℓ containing ϵ and closed under adjacency across facets containing ϵ . Since $\text{st}(\epsilon, K)$ is strongly connected through such facets, $st = \text{st}(\epsilon, K)$. All its cells have split edge ϵ and level ℓ (consistency, level homogeneity), so the final line is a single stellar subdivision $(\epsilon, v)_k$ and the mesh remains conforming.

The new adjacent pairs are of three kinds:

- pairs of children of the same cell, covered by Lemma 4.3;
- pairs of children of two cells of st , covered by case 3 of Lemma 4.4;
- pairs formed by a child and a cell $\sigma' \notin st$. Here σ' is adjacent across a facet $\partial_p c$ with $p \in \{0, \text{type}(c)\}$, so $\text{level}(\sigma') \in \{\ell, \ell + 1\}$ by Lemma 4.2; this is covered by cases 1, 2, 4–7 of Lemma 4.4.

Finally, the weld vertex of a child sits at position $l = \text{type}(\text{parent})$, while its split edge occupies positions 0 and $\text{type}(\text{child}) \neq l$. The scheme is therefore alternating. $\square$

Remark 4.6 (Geometry). The scheme is purely combinatorial; where the new vertex is placed is a separate choice. With midpoints on Kuhn’s cube, cells of equal type are similar [32]. In Section 7 we place vertices at Fubini–Study geodesic midpoints instead; the combinatorics is unchanged.

4.1 Relation to Traxler and Stevenson

Traxler [43] and Stevenson [42] impose a condition on the initial mesh only, and prove that all uniform refinements are conforming [42, Thm. 4.3]. They use a different convention. A tagged simplex $(x_0, \dots, x_d)_\gamma$ is bisected along $x_0 x_d$, and $T_R = (x_d, x_1, \dots, x_\gamma, x_{d-1}, \dots, x_{\gamma+1}, x_0)_\gamma$ has the same children. Two neighbors are *reflected* if the vertex sequence of T or of T_R agrees with that of T' in all but one position. Stevenson’s condition (b) asks that neighbors be reflected

when the refinement edge of either lies on the common facet, and that their neighboring children be reflected otherwise. At level 0, Traxler's type $\gamma = 0$ corresponds to Maubach's type d with the same vertex order [17, Rem. 4]; in both conventions the refinement edge is the first–last pair.

Proposition 4.7. *Let K be a level-0 mesh satisfying (MaubachA), and σ, ω adjacent.*

- (a) *If $\partial_i \sigma = \partial_i \omega$, their vertex sequences agree in all but position i .*
- (b) *If $\partial_0 \sigma = \partial_d \omega$, neither refinement edge lies on the common facet, and the children containing the common facet have vertex sequences that agree in all but one position.*

Hence (MaubachA) at level 0 implies condition (b) of Stevenson, using T and never T_R .

Proof. (a) is immediate. For (b), write $\sigma = \langle a, f_1, \dots, f_d \rangle$ and $\omega = \langle f_1, \dots, f_d, b \rangle$. The refinement edges $\{a, f_d\}$ and $\{f_1, b\}$ do not lie in $\{f_1, \dots, f_d\}$. The children containing the common facet are $c_1 \sigma = \langle f_1, \dots, f_d, v_\sigma \rangle$ and $c_0 \omega = \langle f_1, \dots, f_d, v_\omega \rangle$, which differ only in the last position. $\square$

We do not know whether the converse holds: Stevenson's condition may relate neighbors through T_R , i.e. with their common vertices in reversed order, which equalities of ordered faces cannot express. By Stevenson's Remark 4.4 his condition is necessary for conformity of all uniform refinements, and by Theorem 4.5 (MaubachA) at level 0 is sufficient. What Definition 4.1 adds to these initial-mesh conditions is heredity: it holds at every level, which is what makes the local, inductive proof of Theorem 4.5 possible.

5 Initial meshes: strongly compatible seeds are balanced

Theorem 4.5 moves the whole difficulty to the initial mesh: once the Maubach invariants hold at level 0, refinement takes care of itself. For products of simplices, such as Kuhn's cube, they hold by construction; in general they do not. Our motivating application raises the question for a seed that triangulates a closed 4-manifold, $\mathbb{CP}^2$, rather than a domain of $\mathbb{R}^d$. This section recalls that the strong form of the invariant is balance, and uses this to bring in what is known about balanced triangulations of closed manifolds.

5.1 The equivalence

Definition 5.1. A pure d -dimensional complex K is *balanced* if there is a map $\mathfrak{c} : V \rightarrow \{0, \dots, d\}$, a *coloring*, that is injective on the vertices of every cell. Equivalently, its 1-skeleton is properly $(d+1)$ -colorable.

The next proposition is a reformulation of Traxler's reflected condition [43]. It is stated in terms of colorings in [17, §3.1]. We include a proof in the present language.

Proposition 5.2. *A conforming d -mesh admits a vertex order in every cell making it strongly compatible if and only if it is balanced. The order of each cell is then the order of the colors of its vertices, which is unique up to a global permutation of the colors.*

Proof. ($\Leftarrow$) Order each cell by color. The common facet of two adjacent cells carries every color except the one omitted, which occupies the same position i in both cells, so $i = j$. ($\Rightarrow$) If $\partial_i \sigma = \partial_i \omega$ as ordered simplices, every vertex of the common facet occupies the same position in σ and in ω . The star of a vertex v is strongly connected through facets containing v , so the position of v is the same in all cells containing it. This position defines $\mathfrak{c}(v)$, and it is injective on each cell. Uniqueness follows by the same propagation argument. $\square$

Remark 5.3 (Index-only matching is not enough). A weaker, tempting condition asks only that the omitted vertex sit at the same local *index* on both sides of each facet, i.e. a proper $(d+1)$ -edge-coloring of the dual graph in which every cell sees every color. It does not suffice. In dimension 3, take $\sigma = \langle a, b, c, d \rangle$ and $\omega = \langle d, x, c, a \rangle$, sharing $\{a, c, d\}$ at index 1 on both sides, with common split edge $\{a, d\}$. After one bisection with new vertex m , the children $\langle a, b, c, m \rangle$ and $\langle x, c, a, m \rangle$ share $\{a, c, m\}$ at indices 1 and 0, at type 2. This violates (MaubachA): the split edge $\{a, c\}$ of the first lies in the second, whose split edge is $\{x, a\}$. Index-only matchings are much more common than balance. By König's theorem they exist whenever the dual graph is bipartite, and exhaustive search finds one on Kühnel's $\mathbb{CP}_9^2$, whose dual graph is not bipartite and which is not balanced (Proposition 5.10).

5.2 Classical constructions

Corollary 5.4 ([36, Thm. 9]). *The staircase triangulation of a product of simplices $\Delta^{m_1} \times \dots \times \Delta^{m_k}$ is balanced, with $\mathbf{c}(v_L) = |L|$. In particular Kuhn's triangulation of the cube is strongly compatible.*

Corollary 5.5 ([36, Thm. 10]). *The barycentric subdivision of any mesh is balanced, with the barycenter of δ colored $\dim \delta$. It is produced by the ordered stellar subdivisions $(\delta, v_\delta)_{\dim \delta}$, in order of decreasing dimension.*

Proposition 5.6. *If K_1, K_2 are balanced with colorings $\mathbf{c}_1, \mathbf{c}_2$, the staircase triangulation of $K_1 \times K_2$, built prism by prism from monotone lattice paths, is balanced with $\mathbf{c}(a, b) = \mathbf{c}_1(a) + \mathbf{c}_2(b)$.*

Proof. A cell of the prism $\sigma_1 \times \sigma_2$ is a monotone path $L_0, \dots, L_{m_1+m_2}$ with $|L_p| = p$. Its p -th vertex is $(\sigma_1[L_p^1], \sigma_2[L_p^2])$, of color $L_p^1 + L_p^2 = p$. The coloring is defined on vertices, so it is consistent across prisms. $\square$

For instance, the octahedron is balanced (color $\pm e_k$ by k), so the staircase triangulation of $S^2 \times S^2$ as a product of two octahedra is a strongly compatible seed.

Stevenson's pre-refinement [42, App. A], after Kossaczky, is essentially the barycentric subdivision without the edge-midpoint level: it has $\frac{1}{2}(d+1)!$ cells per simplex instead of $(d+1)!$, and satisfies the weaker matching condition. The coloring initialization of [17] avoids pre-refinement altogether; for balanced meshes it coincides with Proposition 5.2, and for the others it uses more than $d+1$ colors.

5.3 Closed manifolds

For a closed d -manifold M , Dynnikov and Novikov consider three classes of triangulations, which Gaifullin [23, §2] denotes $T_{\text{colour}}(M) \subseteq T_{\text{even}}(M)$ and $T_{\text{bw}}(M) \subseteq T_{\text{even}}(M)$:

- $T_{\text{colour}}(M)$ consists of the balanced triangulations;
- $T_{\text{bw}}(M)$ consists of the triangulations whose top simplices admit a *chess coloring*, i.e. whose dual graph is bipartite;
- $T_{\text{even}}(M)$ consists of the triangulations in which every $(d-2)$ -simplex lies in an even number of top simplices.

Gaifullin shows that $T_{\text{colour}} \subseteq T_{\text{bw}}$ when M is orientable, and that all three classes coincide when M is simply connected. The first inclusion has a one-line proof in our language:

Lemma 5.7 ([23, §2]). *The dual graph of an orientable, strongly connected, balanced, closed pseudomanifold is bipartite.*

Proof. Fix a coherent orientation. Let $\varepsilon(\sigma) = \pm 1$ record whether the color order of σ is positively oriented. Two adjacent cells induce on their common facet the orientations $(-1)^k \varepsilon(\sigma)$ and $(-1)^k \varepsilon(\omega)$ relative to its color order, where k is the missing color. Coherence makes these opposite, so $\varepsilon(\sigma) = -\varepsilon(\omega)$. $\square$

Combined with Proposition 5.2, this identifies the seeds that our framework accepts.

Corollary 5.8. *Let K be a triangulation of a closed combinatorial d -manifold M , $d \geq 2$. Then K is a strongly compatible seed for Maubach bisection if and only if $K \in T_{\text{colour}}(M)$. If M is simply connected, this holds if and only if every $(d-2)$ -simplex of K has even degree.*

The degree criterion is Traxler’s condition for simply connected domains [43], now for closed manifolds. Joswig’s projectivity group gives the general criterion [26]. A lower bound on the size of such seeds is also known:

Proposition 5.9 ([23, Prop. 2.3]). *A balanced combinatorial triangulation of a closed d -manifold with fewer than $3d + 3$ vertices triangulates a sphere.*

5.4 Seeds for $\mathbb{CP}^2$

$\mathbb{CP}^2$ has two classical small triangulations. Kühnel’s $\mathbb{CP}_9^2$ [29] has 9 vertices and is vertex-minimal. Gaifullin’s X [23] has 15 vertices, f -vector $(15, 90, 240, 270, 108)$, and automorphism group $S_4 \times S_3$, which acts by Fubini–Study isometries in Gaifullin’s explicit realization.

Proposition 5.10. *Kühnel’s $\mathbb{CP}_9^2$ is not a strongly compatible seed.*

Proof. $\mathbb{CP}_9^2$ is 3-neighborly, so its graph is K_9 , which is not 5-colorable. Alternatively, it has only $9 < 15$ vertices, and Proposition 5.9 applies. $\square$

Theorem 5.11 (Gaifullin [23]). *X is balanced: color the three vertices of $V_4 \setminus \{e\}$ with 0 and the vertices (a, b) with a , for $a = 1, \dots, 4$. It is a vertex-minimal balanced triangulation of $\mathbb{CP}^2$.*

Corollary 5.12. *Every strongly compatible seed of $\mathbb{CP}^2$ has at least 15 vertices, and Gaifullin’s X , ordered by its coloring, is a strongly compatible seed with exactly 15.*

Proof. Combine Corollary 5.8, Proposition 5.9 and Theorem 5.11. $\square$

Remark 5.13. We verified these facts independently on the facet list of X (with `simpcomp` [21]):

- every vertex has exactly two non-neighbors, namely the other members of its color class;
- all 240 triangles have degree 4 or 6;
- in the color order, all 270 shared facets satisfy $i = j$.

The 108 seed cells are also geometrically uniform: the max/min Fubini–Study edge ratio is 1.4636 in every cell, as expected from the isometric symmetry. By Corollary 5.5, the barycentric subdivision of $\mathbb{CP}_9^2$ is balanced too. In Kühnel’s standard realization, however, several flag barycenters coincide with existing vertices, which produces degenerate cells.

6 Pointerless codes on balanced seeds

6.1 Why codes

An explicit mesh stores each simplex with references to its faces and cofaces. The number of faces per cell grows quickly with the dimension. In the Freudenthal–Kuhn triangulation of $\mathbb{R}^d$, which is the combinatorial type of uniform Maubach refinements of the cube, a k -simplex is determined up to translation by an ordered sequence of k disjoint nonempty subsets of $\{1, \dots, d\}$. There

are therefore $k! S(d+1, k+1)/d!$ of them per d -cell, where S denotes Stirling numbers of the second kind. A full incidence structure storing only the boundary references ($k+1$ integers per k -simplex) already needs:

d	2	3	4	5	6	7	8
simplices of dim. $1..d$ per cell	2.5	4.2	6.2	9.0	13.0	18.8	27.1
boundary references per cell	6.0	12.3	22.5	39.0	65.7	108.3	175.8
LPT code bits (40 levels)	49	52	55	59	62	66	70

A *compact* explicit structure stores cells only, each with its $d+1$ vertex identifiers and $d+1$ neighbor identifiers, i.e. $8(d+1)$ bytes per cell with 32-bit integers (40 bytes for $d=4$) plus bookkeeping for levels and the hierarchy. An LPT code needs about $L + \log_2(2^d d!) + O(\log d)$ bits for L levels, whatever the dimension.

We measured one full incidence structure, VGTL's `nmt<4>` [35], against the pointerless leaf table of Section 6.5, on uniform refinement of Gaifullin's seed with no coordinates and no other data. Both hold exactly the same $108 \cdot 2^D$ leaves.

D (leaves)	<code>nmt<4></code>		pointerless	
	cells stored	bytes/leaf	table	bytes/leaf
8 (27 648)	55 188	222	0.5 MB	19.5
12 (442 368)	884 628	222	8.6 MB	19.5

`nmt<4>` keeps every cell ever created, so that coarsening remains possible, together with all faces; this costs 222 bytes per leaf. The pointerless table stores one 64-bit code per leaf at load factor 0.41, i.e. 19.5 bytes per leaf. That is about $11\times$ less than this full incidence structure, and about $2\times$ less than the 40-byte compact structure estimated above. Neither figure includes per-vertex data such as coordinates, which both approaches need when geometry is required. Beyond memory, the main benefit of the codes is that neighbors, children and parents are computed rather than stored. This keeps the data structure trivially consistent under refinement.

6.2 LPT codes

Atalay and Mount [5] encode the Maubach hierarchy of a Kuhn-triangulated cube $[-1, 1]^d$. Its reference root simplex is

$$S_\emptyset = \{x \in \mathbb{R}^d : 1 \geq x_1 \geq x_2 \geq \dots \geq x_d \geq -1\},$$

whose facet H_0 lies on $x_1 = 1$, facet H_j ($0 < j < d$) on $x_j = x_{j+1}$, and facet H_d on $x_d = -1$. A descendant is encoded by its *LPT code* (ℓ, Π, Φ) :

- ℓ is the level modulo d ;
- $\Pi = (\pi_1, \dots, \pi_d)$ is a signed permutation;
- Φ is the orthant list. After every d levels, one bit is appended to each coordinate. We read these bits as integers $o_1, \dots, o_d \in [0, 2^r)$, where r is the number of completed rounds and the oldest bit is the most significant.

Their Theorem 1 gives the vertices explicitly. With $T_j = 2^r - 1 - 2o_j$, the k -th vertex is

$$x^{(k)} = \frac{T + y^{(k)}}{2^r}, \quad y_{|\pi_j|}^{(k)} = \text{sgn}(\pi_j) v_{kj}, \quad v_{kj} = \begin{cases} 1, & k \geq j, \\ 0, & k < \min(j, \ell), \\ -1, & \ell \leq k < j, \end{cases} \quad (2)$$

for $k = 0, \dots, d$ and $j = 1, \dots, d$. Here $y^{(k)} \in \{-1, 0, 1\}^d$ depends only on (ℓ, Π, k) , while T depends only on the orthant list. Their Theorem 2 computes the code of a facet neighbor by table lookup. It describes neighbors in the infinite Kuhn arrangement: in the cube, a neighbor outside one root simplex is another root simplex of the same cube. Our implementation descends from the one we wrote for the Adaptive Voids algorithm [20], where LPT codes over the six Kuhn tetrahedra of a cube drive an adaptive subdivision of a solid.

6.3 Balanced seeds

Let K_0 be a balanced d -dimensional seed with N_0 cells, each ordered by color, and identify every seed cell $\tau = \langle \tau[0], \dots, \tau[d] \rangle$ with $S_\varnothing$ by sending the k -th vertex of $S_\varnothing$ to $\tau[k]$. A cell of the refined seed is a pair (s, κ) : a seed index s and an LPT code κ of the reference hierarchy.

We work *combinatorially*: a vertex of the hierarchy of τ_s is identified with its barycentric coordinates $w \in \Delta^d$ with respect to τ_s . These are dyadic rationals, and by (2) the coordinates $w(\kappa, m)$ of the m -th vertex of (s, κ) depend on κ only. A vertex on a facet of τ_s is also a vertex of the neighboring seed cell, via the identification of the common facet. A geometric realization is applied afterwards: affine, or by Fubini–Study midpoints as in Section 7. Since it is defined edge by edge from the endpoints of the split edge, it respects these identifications.

Proposition 6.1 (seed-boundary test). *Facet ∂_i of the cell with code $\kappa = (\ell, \Pi, \Phi)$ lies on the facet H_j of its seed cell if and only if:*

- (i) *the linear part of the equation of H_j is constant on $\{y^{(k)} : k \neq i\}$, i.e. $y_1^{(k)} \equiv 1$ ($j = 0$), $y_{j+1}^{(k)} - y_j^{(k)} \equiv 2\delta$ with $\delta \in \{-1, 0, 1\}$ ($0 < j < d$), or $y_d^{(k)} \equiv -1$ ($j = d$); and*
- (ii) *$o_1 = 0$ ($j = 0$), $o_{j+1} - o_j = \delta$ ($0 < j < d$), or $o_d = 2^r - 1$ ($j = d$).*

At most one j satisfies (i), which depends only on (ℓ, Π, i) .

Proof. Substitute (2) into the equation of H_j . For $0 < j < d$, it reads $T_j + y_j^{(k)} = T_{j+1} + y_{j+1}^{(k)}$ for all $k \neq i$. Since T does not depend on k , this forces $y_{j+1}^{(k)} - y_j^{(k)}$ to be constant, equal to $T_j - T_{j+1} = 2(o_{j+1} - o_j)$. For $j = 0$, the equation $T_1 + y_1^{(k)} = 2^r$ gives $2o_1 = y_1^{(k)} - 1$; since $o_1 \geq 0$ and $y_1^{(k)} \leq 1$, this means $o_1 = 0$ and $y_1^{(k)} = 1$. The case $j = d$ is symmetric. At most one j qualifies, because the d vertices of a facet span a hyperplane with a unique normal direction, and the facet normals of a simplex are pairwise non-parallel. $\square$

Condition (i) is precomputed in a table indexed by (ℓ, Π, i) that stores j and δ , or “none”. For $d = 4$ the table has $4 \cdot 384 \cdot 5 = 7680$ one-byte entries. The test is then a single lookup followed by a single integer comparison.

Proposition 6.2 (seed crossing). *Suppose facet ∂_i of $c = (s, \kappa)$ lies on the facet $\partial_j \tau_s$ of its seed cell, and let s' be the seed cell sharing that facet. Then the neighbor of c across $\partial_i c$ is $c' = (s', \kappa)$, the same code in the other seed cell, and $\partial_i c' = \partial_i c$ as ordered simplices.*

Proof. By Proposition 5.2, $\partial_j \tau_s = \partial_j \tau_{s'}$ as ordered simplices, i.e. $\tau_s[p] = \tau_{s'}[p]$ for every $p \neq j$. For $m \neq i$, the m -th vertex of c lies on $\partial_j \tau_s$, so its coordinates satisfy $w_j(\kappa, m) = 0$. The m -th vertex of c' has the same coordinates with respect to $\tau_{s'}$, hence the same position on the common facet. It is therefore the same vertex of the refined seed. So c' contains $\partial_i c$, in the same positions, has the same level as c , and lies in the other seed cell. Uniform refinements of a strongly compatible mesh are conforming (Theorem 4.5, applied to every cell of a level in turn), so exactly two cells of that level contain $\partial_i c$. Hence c' is the neighbor. $\square$

This is where balance, rather than any compatible ordering, is used: without it the two identifications would differ on the common facet by a permutation, and the code would have to be transformed.

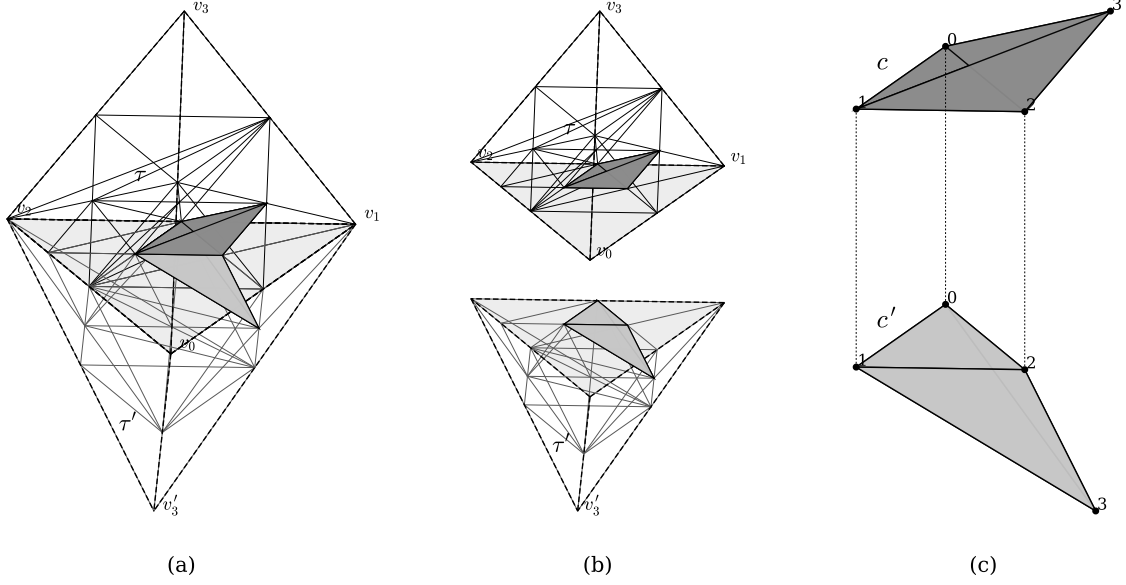


Figure 1: Propositions 6.1 and 6.2 in dimension 3. (a) Two color-ordered seed tetrahedra τ and τ' , glued along their common facet $\langle v_0, v_1, v_2 \rangle$ (light gray), after adaptive Maubach bisection towards a point just above that facet. The conforming closure propagates across the seed boundary. Dashed lines are seed edges. (b) The same mesh, exploded along the normal of the common facet: both seed cells induce the same subdivision on it. (c) The leaf $c \subset \tau$, whose facet $\partial_3 c$ lies on the seed boundary, and its neighbor $c' \subset \tau'$. Both have the same bisection code (path 010010 from their roots), and the shared facet occupies local positions 0, 1, 2 in both. The figure is generated by a script that asserts both facts.

6.4 Neighbor finding

To find the neighbor of $c = (s, \kappa)$ across $\partial_i c$:

1. *Sibling facet.* If $\partial_i c$ is the facet created by the last bisection ($i = d$ for a 0-child, $i = \text{type}$ for a 1-child), return the other child of the parent (Lemma 4.3).
2. *Boundary test.* Otherwise, look up (j, δ) for (ℓ, Π, i) and check the orthant condition of Proposition 6.1.
3. *Interior facet.* If the test fails, return (s, κ') with κ' given by Atalay and Mount's Theorem 2.
4. *Seed crossing.* Otherwise return $(\text{adj}[s][j], \kappa)$ (Proposition 6.2), where adj is the $N_0 \times (d+1)$ seed adjacency table. Root cells use this rule directly, with $j = i$.

Rule 2 is indispensable. Across a seed-boundary facet, Atalay and Mount's formula returns a cell of the reflected reference simplex, i.e. one with negative barycentric weights. Such crossings are frequent: about 24% of the non-sibling facets sampled at realistic depths lie on the seed boundary. Figure 1 illustrates rules 2 and 4.

Verification. We checked every returned neighbor *exactly* on Gaifullin's seed ($d = 4$). Both cells were written as integer barycentric coordinates over the 15 global vertices, with common denominator 2^{11} , and we checked that the returned cell is a different cell of the same level containing the queried facet. The test covered 10^6 random cells over the whole representable depth range and five million queries: 977 070 sibling, 3 317 979 interior and 704 951 seed-crossing. None failed, and away from sibling facets the facet always sat at the same local index on both

sides. An earlier implementation computed the boundary test from the full matrix of barycentric weights ($O(d^2 + rd)$ integer operations) and resolved crossings by floating-point point location. The table-based test agrees with it on 7.8 million facets, and same-code crossing agrees on 1.4 million crossings. The new rules take 4.3 ns and 5 ns per query, against 79 ns and 211 ns.

6.5 The leaf set

The mesh is the set of codes of the current leaves, stored in an open-addressing hash table: one 64-bit slot per leaf, linear probing, and backward-shift deletion, which avoids tombstones under the erase-two-insert churn of bisection. For Gaifullin’s seed a code fits in one word: 7 bits of seed index, 2 bits for ℓ , 9 for Π , 4 for r and 4×10 for the orthant list, i.e. 62 bits and a depth of up to 43. Two operations are built on the table:

- `compat_bisect( $c$ )` is Algorithm 2 on codes. It bisects coarser neighbors recursively before bisecting c , so the mesh stays graded: no facet separates cells more than one level apart;
- `neighbor_leaf( $c, i$ )` walks up the ancestor chain of the combinatorial neighbor to the current leaf, or reports that the region has been refined further.

An exhaustive graded-invariant test, with about 40 000 cascading bisections, 4 million leaves and 11.6 million facet checks, found no violation. Vertex identifiers are not part of the code. When needed (for geometry), a new vertex receives an integer identifier, keyed by the identifiers of the endpoints of the split edge in a separate hash table; vertices are about $1/d!$ as numerous as cells.

Limitations. Enumerating the several finer leaves that border a coarser facet is not implemented; such queries are reported, never guessed. Seeds that are compatible only in the weak sense of (MaubachA) would need a permutation correction in Proposition 6.2. For the initialization of [17] with $N > d$ colors, a cell is a face of a virtual N -simplex. Refinements of the virtual extension do not restrict to refinements of the mesh [17, Rem. 12], so it is unclear how to encode them.

7 Example: plane curves inside $\mathbb{CP}^2$

Let $F(X, Y, Z)$ be homogeneous of degree n and $C = \{F = 0\} \subset \mathbb{CP}^2$ smooth. Then C is a compact Riemann surface of genus $(n-1)(n-2)/2$, and a real surface of codimension 2 in $\mathbb{CP}^2$. We triangulate it as follows.

Mesh of $\mathbb{CP}^2$. The seed is Gaifullin’s X with his coordinates [23], moved by a random unitary change of basis of $\mathbb{C}^3$ to avoid special alignments between the curve and the seed. It is refined by pointerless Maubach bisection (Section 6). Each new vertex is placed at the Fubini–Study geodesic midpoint of the split edge, after aligning the phases of the two homogeneous representatives. Coordinates are cached by vertex identifier.

Extraction. For a leaf c , each of its 10 triangular 2-faces is intersected with C by Newton’s method in the best-conditioned affine chart of that face. Results are cached by the face’s vertex identifiers, so adjacent cells see identical crossings. On each tetrahedral facet, the crossings are paired into arcs, by minimal total geodesic length when there are four. The arcs are then assembled into closed polygons approximating $C \cap c$. A cell is *bad* when this assembly fails, e.g. at a node of degree $\neq 2$, which happens near tangencies.

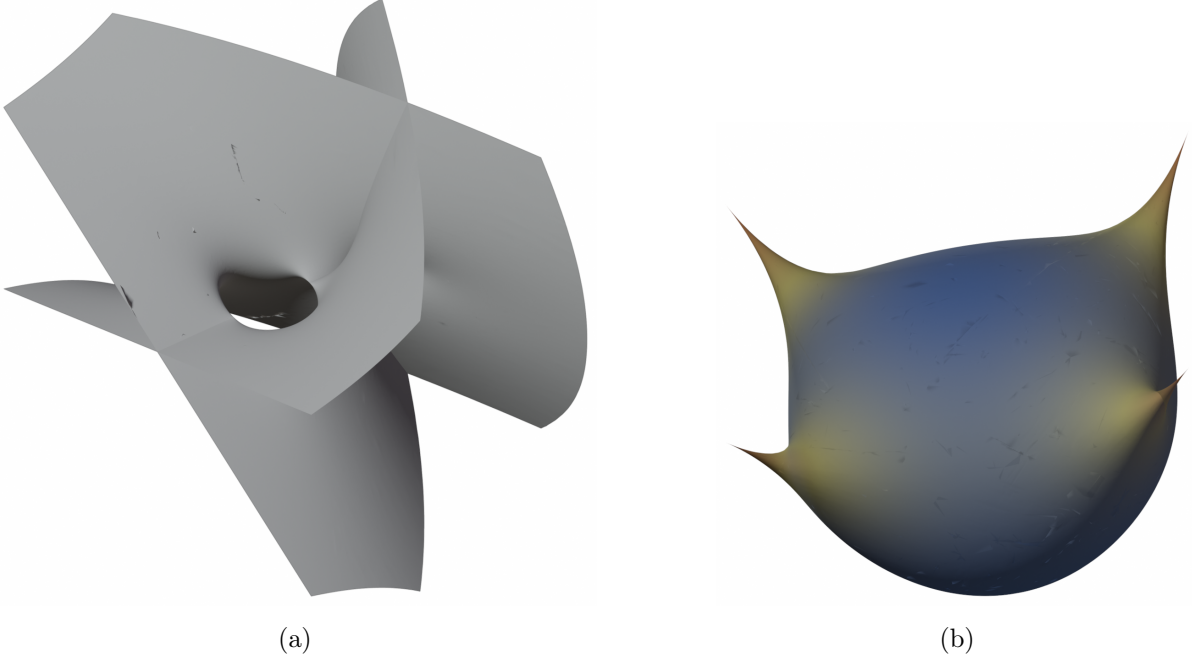


Figure 2: Curves triangulated inside $\mathbb{CP}^2$. (a) The elliptic curve $X^2Z - Y^3 + YZ^2 = 0$, i.e. $w^2 = z^3 - z$ in the chart $Z = 1$ ($w = X$, $z = Y$), displayed as the graph $(\operatorname{Re} z, \operatorname{Im} z, \operatorname{Re} w)$, clipped to $|\operatorname{Re} z|, |\operatorname{Im} z| \leq 2$ and $|\operatorname{Re} w| \leq 3$; depth 18, 0.39% bad cells. (b) The Fermat quartic $X^4 + Y^4 + Z^4 = 0$ (genus 3) under Dutter’s chart-free map $\tilde{\alpha} : \mathbb{CP}^2 \rightarrow \mathbb{R}^3$ [19], colored by distance to the center and drawn semi-transparent; depth 17, 3.35% bad cells. Both surfaces were closed as described in Section 7. The faint marks are discontinuities of the interpolated normals, at the closing cones (most marks in (b)) and at thin polygons of the extraction (the darker streaks in (a)), not holes.

Continuation. Starting from one seed cell that meets C , a frontier of cells is processed as follows:

- a cell that meets C and is below the target depth is bisected with `compat_bisect`, and its new leaves join the frontier;
- a cell at the target depth is extracted, and its neighbors across the five facets, found by `neighbor_leaf`, join the frontier.

Moving through the mesh thus uses only the combinatorial operations of Section 6. Two optional mechanisms refine further near tangencies:

- a *tangency threshold* on a Cauchy–Riemann transversality score computed at each crossing;
- *repair rounds* that re-bisect bad cells.

The continuation reaches every cell whose 2-faces are crossed by a connected curve. A small closed component of $C \cap c$ that meets no 2-face of c would be missed, and a reducible curve needs one starting cell per component.

Topological consistency. The output is not certified. Its topology can nevertheless be checked, directly on the extracted 2-complex in $\mathbb{CP}^2$, independently of any projection. Vertices are identified by the mesh vertex, edge or 2-face on which the crossing lies. Bad cells leave holes. We split each *pinched* vertex into one copy per component of its link. In all our runs these vertices lie on hole boundaries, where a hole touches itself or two holes meet. If the holes

Table 1: Robustness across five random rotations of the seed (depth 14, tangency threshold 0.05, two repair rounds). “Closed” reports χ of the closed surface, which in all runs has no boundary and one component.

curve	bad cells (min–max)	genus estimate	closed χ	expected
elliptic ($g = 1$)	1.00–1.90%	1 in 5/5	0 in 5/5	0
Fermat quartic ($g = 3$)	6.03–7.21%	3 in 5/5	–4 in 5/5	–4

are discs, then $g = (2c - b - \chi)/2$, where c is the number of components and b the number of boundary loops. A closing step then turns the complex into a closed surface:

- it keeps the main component;
- it caps each hole by a cone from a new apex, placed at the loop’s Fubini–Study barycenter and moved onto C by Newton’s method;
- it checks that the result is a connected closed surface with $\chi = 2 - 2g$.

These are necessary conditions. A missing handle and a spurious one would cancel in χ , and χ says nothing about the embedding. They are nevertheless a useful sanity check for an uncertified method. Table 1 repeats the construction for five random rotations of the seed. In all ten runs the genus estimate is exact, and the closed surface passes every check. The bad-cell rate varies moderately with the rotation. Section A gives further measurements: the bad-cell rate against depth, a breakdown of the causes of failure, and the accuracy of the extracted vertices.

8 Conclusion

Stellar subdivision schemes reduce the correctness of Maubach bisection to three local lemmas about ordered stellar operations. On conforming meshes, the strong form of the invariant is balance. This connects the choice of a seed with the classes of triangulations studied by Dynnikov, Novikov and Gaifullin. For simply connected closed manifolds, a triangulation is a strongly compatible seed exactly when all its codimension-2 faces have even degree, and for $\mathbb{CP}^2$ Gaifullin’s triangulation is the smallest such seed. Balance is also what lets pointerless LPT codes run over arbitrary seeds, with a table-driven, floating-point-free neighbor procedure.

Several questions remain open:

- whether Stevenson’s matching condition implies (MaubachA) after a suitable choice between T and T_R ;
- pointerless codes for weakly compatible seeds and for the $N > d$ initializations of [17];
- small balanced seeds for other closed manifolds, for instance via cross-flips [25, 46];
- certification of the example’s extraction.

Note on AI assistance

The implementation described here was developed with Claude (Anthropic) acting as a coding assistant, under the author’s direction, review and testing. This manuscript was likewise prepared with Claude’s assistance.

Code availability. The ordered-mesh data structures, the stellar subdivision algorithms and the example programs are part of VGTL [35]. The LPT codes and their extension to balanced seeds, with the test suites of Section 6, are in the `lpt` repository [34]. Scripts regenerating all figures and tables are in VGTL’s `report/article/figures`.

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A Further measurements for the example

All runs below use one fixed random rotation of the seed (generator seed 12345).

Bad-cell rate versus depth. Figure 3 shows the bad-cell rate for four curves of degree 2 to 4. It decreases steadily, roughly halving every two levels. The tangency threshold combined with repair rounds lowers it by a factor of 1.5 to 3 at every depth. Curves of higher degree need a few more levels to reach the same rate. For the elliptic curve:

depth	mode	leaves	good cells	bad cells	bad rate
14	continuation only	327 370	27 397	887	3.14%
14	+ repair rounds (2)	327 370	30 605	730	2.33%
14	+ tangency threshold 0.05	612 978	42 278	904	2.09%
14	+ both	612 978	45 709	729	1.57%
20	continuation only	4 620 102	222 981	1 348	0.601%
20	+ repair rounds (2)	4 678 426	226 677	678	0.298%
20	+ tangency threshold 0.05	6 788 844	335 517	1 260	0.374%
20	+ both	6 842 524	339 147	610	0.180%

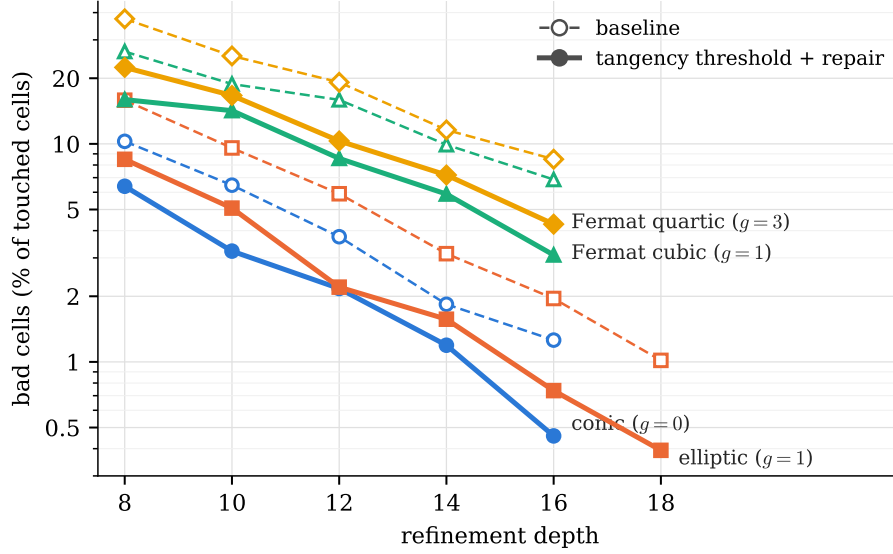


Figure 3: Percentage of touched cells whose extraction fails, against the refinement depth. Dashed lines with open markers: continuation alone. Solid lines: with tangency threshold 0.05 and two repair rounds.

Causes of failure. We instrumented the extraction at depth 14 (136 786 touched cells over six curves, 8.0% bad):

- 82% of the bad cells fail on node degree, and 18% on facets with three crossings;
- cells with a facet touched at a single point are seven times more likely to be bad;
- the transversality score separates good cells (median 0.21) from bad ones (median 0.12), with a monotone gradient in bad rate across score buckets.

Fubini–Study midpoints split cell volumes unevenly, but this imbalance does not correlate with bad cells.

Genus of the raw extraction. With the notation of Section 7:

curve	depth	bad cells	pinched	c	b	χ	g est.	g
conic $XY - Z^2$	14	367 (1.2%)	3	2	41	-37	0	0
elliptic	14	729 (1.6%)	8	4	84	-78	1	1
elliptic, Figure 2(a)	18	701 (0.39%)	3	2	104	-102	1	1
Fermat cubic	14	2 438 (5.9%)	35	8	253	-239	1	1
Fermat quartic	17	5 388 (3.4%)	36	11	656	-640	3	3

In every case the closed surface is connected, has no boundary, and has $\chi = 2 - 2g$. The islands dropped by the closing step have 1 to 18 faces, and Newton’s method moved every cone apex onto the curve.

Accuracy. At depth 20 we evaluated $|F|$, with unit-norm homogeneous representatives, at the 164 390 distinct extracted vertices. The median is 1.0×10^{-16} , the 99th percentile 4.3×10^{-13} and the maximum 9.8×10^{-13} . This measures the vertices only; the distance between the polygons and the curve was not measured.

Mesh size. Continuation recovers the same leaf set and bad cells as a global scan of all 108 seed hierarchies with the same test, on six curves at depths 8, 12 and 14. It yields meshes 3–6 times smaller at depth 14 than an earlier criterion that refines all of $\mathbb{CP}^2$. The pointerless mesh reached depth 20, with 6.8 million leaves, on a machine with 7.2 GB of RAM, where the `nmt<4>`-based implementation had peaked at 4–5 GB around depth 18.